Access Is Not Use: Auditing Causal Bypass of Force in Behavior-Cloned Vision-Language-Action Policies

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Abstract*—Vision-language-action (VLA) policies increasingly receive physical condition signals—wrist wrench, contact, vacuum-seal state—yet whether a trained policy *causally uses* them is unmeasured, and standard validation cannot tell: we show that a policy that uses force and one that routes around it can train to validation losses equal in the fourth decimal place. On a real-robot suction case-picking task where the cases carry battery cells—a $15\,\mathrm{N}$ contact-force limit, and a critical transition (stop descending on contact) observable only through force—a naively fine-tuned VLA receiving the full wrench presses through contact in every trial and never stops on its own. We present a recipe that makes the policy use force where it matters: a condition vector $\hat{c}$ computed from signals the policy already receives, injected through FiLM with the raw wrench masked—a grounded pathway that adds routing, not information—trained on press-retreat demonstrations that press to a varied force target and retreat before the seal. Forcing $\hat{c}$ at inference measures its causal authority, and a controlled ablation shows each ingredient is necessary: unmasked, the pathway trains causally dead (*bypass*); ungrounded, the policy is the worst in the study. The naive policy's force response is a template that fades beyond the demonstrated range; the conditioned policy's grows monotonically, stopping and retreating at forces beyond its training data. On the robot, every conditioned pick stops on its own—median contact force $2.5\,\mathrm{N}$ across three stack heights—where the naive policy is externally aborted in every attempt, and live counterfactual probes reproduce the offline crossover on the physical system. [TODO: scale sentence ($\pi_{0.5}$/pi0) pending the offline re-run]

***Index Terms*—imitation learning, vision-language-action models, force control, causal confusion, contact-rich manipulation**

I. Introduction

Behavior-cloned vision-language-action (VLA) policies are rapidly acquiring physical condition signals. Wrist force/torque, contact events, and task-state bits now routinely appear in the observation vector, and a growing line of work fuses force into the policy backbone for contact-rich manipulation [1]–[4]. All of this work shares a validation recipe—add the signal, retrain, and validate by task success (occasionally alongside rollout force statistics [2], [3])—and none of it measures whether the trained policy *causally uses* the signal when it matters: whether, holding everything else fixed, changing the force input changes the action. Imitation learning offers no such guarantee. The objective asks the policy to reproduce *what* the expert did, not *why*: a clone can imitate the behavior convincingly while being driven by entirely different cues than the expert was. Imitation pins down the trajectory, not the mechanism—equally faithful clones can act from opposite causes, and the loss cannot tell them apart [5], [6].

The consequence is easiest to see where force is the only signal that can carry the task. We study real-robot suction case picking with a single head-mounted camera and no wrist camera: the arm must stop descending the moment the suction cup meets the case, and stack heights differing by a few millimeters—the entire success margin—are visually near-indistinguishable at head-camera range. The wrench is in the state the policy receives, and every demonstration exhibits the stop. Yet a naively fine-tuned VLA—full wrench in its state, hereafter the *naive policy*—presses through contact in every evaluation trial and never initiates a stop of its own: each run ends only when an external safety layer, separate from the policy, trips at its $15\,\mathrm{N}$ force limit—set to protect the battery cells the cases carry (15.4–18.7 N measured at abort)—the policy still commanding descent. The task turns on a single transition—stop when force rises—and it is precisely this transition that the policy fails to bind to force.

Our fix changes how force is delivered, not what is delivered. A condition vector $\hat{c}$ (contact, normal force, seal) is *computed* from signals the naive policy already receives and injected through FiLM modulation [7]; the raw wrench is masked from the state so that the same signal does not enter twice, making $\hat{c}$ the *only* pathway from force to action. We call this pathway *grounded* because its content is computed from measured signals, and the policy trained behind it the *conditioned policy*. By construction it receives no information the naive policy lacks, and because the pathway is explicit and unique, forcing $\hat{c}$ at inference directly measures its *causal authority*—the fraction of commanded descent the condition can cancel.

The mask is load-bearing. Leave the raw wrench in the state and the conditioning pathway trains causally dead at unchanged validation loss: training routes force through the already-functional raw path and starves the dedicated one.

We call this *bypass* and exhibit it in Sec. V; it is why a fix of this kind cannot be certified after the fact, and why ours holds by construction. The demonstrations complete the recipe: each press-retreat episode presses to a randomized force target and retreats before any seal event, so the force at which the expert stops varies from episode to episode—in the data, depth, appearance, and timing no longer predict the stop; only the force reading does. The same interface that fixes the policy is the instrument that audits it (Sec. IV). [MEMO: open (08-12): demote "The demonstrations complete the recipe" to setting-the-stage wording, and drop the data co-attribution from the one-sentence summary at the end? Follow-up of the architecture-emphasis decision. Also optional: restore "replicated in two independent model pairs" somewhere in para. 3 — it left the intro with the contributions rework.]

The resulting policy is not uniformly better—it is different exactly where the task is decided. In a matched comparison—same data, same initialization, imitation error indistinguishable—the naive policy *does* respond to force at the demonstrated magnitudes, as strongly as the conditioned policy or more so. But its response is a *template*: sized to the demonstrated forces, it fades or saturates as force rises past them, never strengthening into a stop. The conditioned policy's response instead grows with force, to the point of stopping and backing away at forces beyond anything in the training data (Sec. V). The mechanism carries to the real closed loop: while the naive policy drives contact force past the 15 N safety limit on every attempt, the conditioned policy stops on its own in every pick—across three stack heights, at a median contact force of 2.5 N—and live counterfactual probes reproduce the offline crossover on the physical system (Sec. V-D).

Our contributions:

- **From force runaway to force regulation, with the same demonstrations.** A grounded FiLM pathway—$\hat{c}$ computed from signals the policy already receives, injected through FiLM, the raw wrench masked—turns the same press-retreat demonstrations into a policy that brakes monotonically with force instead of reproducing a saturating template: on the robot it stops under the 15 N abort limit in every pick (median contact force 2.5 N) on a transition the naive policy fails uniformly (Sec. IV, V-D).
- **An instrument for causal authority.** The pathway is measurable by construction: a counterfactual probe battery—condition forcing, state transplants with pathway decomposition, dose-response sweeps, closed-loop press simulation, and on-robot frozen-observation counterfactuals—quantifies moment-resolved causal use and transfers to the real system (Sec. IV-C).
- **Robustness beyond the demonstrations.** The conditioned policy's stop is bound to force itself, not to force's correlates—under intervention, its response runs entirely through the conditioning pathway—and this buys robustness on the two axes we test: braking grows monotonically at forces beyond anything demonstrated, stopping and retreating where the naive policy's template saturates and presses through, and picks succeed $3/3$ at a stack height absent from the demonstrations. The failure class the recipe corrects persists unchanged at 3.6B scale (Sec. V, V-D).

In one sentence: *a policy that stops when it feels what the demonstrations felt and a policy that stops harder the harder it is pressed are indistinguishable in loss and similar at the demonstrated forces—they differ exactly where the task is decided, and only demonstrations in which force alone predicts the stop, behind a grounded, masked FiLM pathway, yield the second.*

II. Related Work

Force- and tactile-conditioned VLA policies. ForceVLA fuses wrist force as dedicated tokens through a mixture-of-experts backbone [1]; TA-VLA injects torque adapters into the action decoder [8]; PhaForce schedules policy phases on contact and force signals [2]; ForceFlow [3], FARM [9], and FILIC [4] condition diffusion or impedance-loop policies on force and tactile input; TacForeSight learns force-conditioned tactile prediction [10]. These works *design for* force use and validate it with task success and retraining ablations, sometimes alongside rollout force statistics [2], [3]. None of these measurements establishes causal use at the decision moment: an ablation compares two differently-trained functions, rollout statistics are observational, and success can be carried by correlates of force rather than force itself. The nearest analyses stop short of the force input: TA-VLA perturbs input tokens of its *baseline* to site the adapters and reports correlational representation similarity [8]; Tactile-VLA's dose-response intervenes on the *instruction*, not the force stream [11]; ForceFlow masks its trained policy's *visual* input and reads out task success [3]. We do not claim these systems bypass their force inputs—their tasks may contain force information that vision cannot substitute, leaving no shortcut to take. Our claim is narrower and falsifiable: *when* the demonstrations correlate the condition with cheaper cues, a policy with a dedicated force pathway can train to indistinguishable loss while routing around that pathway (we exhibit exactly this with our unmasked control), and no metric in the standard validation stack detects it. We contribute the missing measurement: an inference-time intervention on the trained policy's force input, with the action as the readout.

Intermediate condition representations. Injecting a compact, task-level representation between observation and action is a recurring design: RT-Affordance conditions on predicted affordance plans [12], and FiLM modulation [7] is the standard mechanism for conditioning a backbone on side information. Our $\hat{c}$ shares the shape but differs in role: it is *computed* from signals the policy already receives—not a learned head that could itself be bypassed—and it is paired with a mask that removes the raw signal, turning the representation from a convenience into the policy's sole, measurable route to force.

Causal confusion and shortcut learning. That imitators latch onto spurious correlates is well established: causal confusion in IL [5], copycat behavior on observation histories [13], and shortcut learning in deep networks broadly [6],

[14]. Recent work documents shortcuts in generalist robot policies [15] and uses targeted interventions to expose or correct them [16], [17]. The symptoms are on record in force-conditioned policies themselves: FuSe observes fine-tuned generalists ignoring newly added sensors [18], and FM-VLA diagnoses a history-length shortcut in its own force stream and patches it with augmentation—verifying the fix, again, only by task success [19]. We instantiate this literature where it has not been applied—force conditioning in fine-tuned VLAs—and add two elements: a quantitative demonstration that *training loss carries zero signal* about which input the policy is bound to (equal to the fourth decimal in two independent model pairs), and the usage/form dissociation: the data's correlation structure decides *whether* a signal is used, while the conditioning architecture decides *what form* its use takes under extrapolation.

Privileged supervision and interventional evaluation. Privileged-teacher distillation [20], [21] assumes access to a signal worth distilling; our probes answer the prior question of whether the deployed policy uses the signal at all. The closest interventional precedent is ForceSight, which ablates predicted force *goals* at its controller under matched initial conditions [22]; the intervention sits on the output side of a model that takes no force input, and reads out end-of-episode success rather than the action. Our press-retreat prescription is a data-side intervention in the spirit of decorrelation fixes for causal confusion [5], but designed from a physical argument (Sec. VI) that reactive gating is arithmetically insufficient on stiff contact, so the demonstrations must teach anticipatory approach and retreat.

III. Task, Setting, and the Failure

Task and platform. We study suction case picking on a Dexmate Vega-1p, a dual-arm mobile manipulator: the arm descends onto a stack of rigid cases, must stop when the suction cup meets the top case, hold until vacuum seal is confirmed, and lift (Fig. **??**). The cases carry battery cells, which tolerate little compression—over-pressing risks the product, not just the pick—so deployment enforces a 15 N contact-force limit. The margin is tight in both directions: too little press and the seal fails; a few millimeters too much and contact force climbs past the limit, since the stack is stiff ($\sim$5 N/mm, Sec. VI). The policy controls a 7-D end-effector action (3-D position and 3-D rotation increments plus a suction command) at 15 Hz, executed in chunks of 5 actions ($\sim$330 ms between observations).

Observations. The policy observes a single head-mounted camera (RGB and colorized depth; no wrist camera) and a 15-D state: end-effector position and quaternion, suction command, vacuum-seal bit, and the 6-D wrist wrench. The setting is a deliberately clean diagnostic: at head-camera range, case-stack heights differing by millimeters—the entire success margin—are visually near-indistinguishable (Fig. **??**, right), while the wrench and seal bit signal the contact transition directly. Every condition the task depends on is *in the state the policy receives*; the question is whether the policy binds its behavior to it.

Demonstrations. We collect 100 training episodes (19,445 frames) and hold out 6 validation episodes (1,169 frames) with a scripted expert designed to *decorrelate* force from its habitual correlates: each episode descends with suction on, presses to a randomized force target drawn from $[8, 15]$ N, retreats 5–10 mm from the contact plane, then hovers and lifts immediately on seal. Episodes are collected at two different stack heights, so the contact plane itself moves across the dataset. Every episode thus demonstrates the counterfactual the task requires: force rises $\rightarrow$ stop $\rightarrow$ move back up, *before* any seal event, at a force level that varies independently of depth. This is the data-side prescription whose rationale we develop in Sec. VI; here it matters that the naive policy and all conditioned variants are trained on exactly these episodes.

The failure. Deployed on the robot, the naive policy—a SmolVLA [23] fine-tuned on these demonstrations, with the full wrench and seal bit in its state—fails every evaluation trial the same way: it descends, makes contact, and keeps pressing—never initiating a stop of its own—until the external safety layer aborts the run (15.4–18.7 N measured at abort). The transition that the demonstrations exhibit in every single episode, driven by the one signal that observably announces it, is the transition the policy does not perform. The rollout says the policy is bad; it does not say *why*. The policy might ignore force entirely; it might respond to force but too weakly; it might respond in a form that collapses precisely when force exceeds the demonstrated range. These hypotheses have opposite prescriptions (more data? different sensors? different architecture?), and none of them is distinguishable from training curves or rollouts. Distinguishing them requires intervening on the signal itself—the instrument we build next.

IV. An Instrument for Causal Authority

The architectural half of our recipe doubles as an instrument. It has two parts: a conditioning pathway constructed so that interventions on it are meaningful (IV-A–IV-B), and a battery of counterfactual probes that perform the interventions (IV-C). Throughout, the design goal is *measurement validity*, not architectural novelty: every component exists to make one number well-defined—how much does the condition causally move the action?

A. Computed condition vector $\hat{c}$

From the wrench F and seal bit already present in the naive policy's state we compute a small set of task-critical conditions—the physical cues a human operator references ("if you feel contact, stop; if the seal is on, lift"):

$$\begin{aligned} \hat{c}_{\text{contact}} &= \text{clip}((\|F\| - F_0)/\tau,\ 0,\ 1), \\ \hat{c}_{\text{fmag}} &= (\|F\| - F_0)/\tau_m \quad \text{(unsaturated)}, \\ \hat{c}_{f_z} &= (f_z - o_z)/\tau_z, \qquad \hat{c}_{\text{seal}} = \text{seal}, \end{aligned} \tag{1}$$

with offsets and scales calibrated to the measured force statistics of the demonstrations (e.g., F_0 at the contact-onset magni-

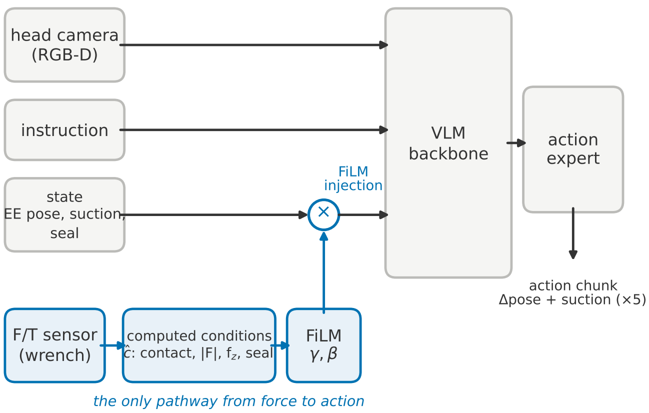

Fig. 1. The conditioned policy. The raw wrench does not enter the policy state; force reaches the action only as the computed condition vector $\hat{c}$, injected through FiLM modulation of the state token entering the VLM backbone. Because this pathway is explicit and unique, forcing $\hat{c}$ at inference measures the causal authority of the condition.

tude).[1] Two properties are essential. *Computed, not learned:* $\hat{c}$ is a fixed function of observed signals, so its physical meaning cannot drift during training—which is what makes forcing $\hat{c}$ at inference a well-defined intervention rather than a perturbation of an arbitrary latent. *Zero new information:* every channel is a deterministic function of state dimensions the naive policy also receives, so conditioning changes *routing*, never *content*—the "it works because it got extra sensing" objection is excluded by construction. The channels are deliberately plural: keeping contact, force magnitude, and seal separate is what later lets the probes attribute authority to specific signals (Sec. V-A).

B. FiLM injection and the force mask

$\hat{c}$ enters the policy through FiLM modulation [7] (initialized to identity), at the state token entering the VLM backbone.[2] The policy still receives the force information—only the structure of its delivery changes. Because $\hat{c}$ is computed *from* the wrench, leaving the raw wrench in the state would hand the policy the same signal twice, once in each form; we therefore mask the wrench dimensions from the state (Fig. 1). Forcing $\hat{c}$ at inference then manipulates *all* of the force information the policy receives: if the policy responds to force at all, it must show in how the action changes with $\hat{c}$—the quantity the probes below measure. [MEMO: revive in Sec. VII if useful: the mask strictly *reduces* the conditioned policy's information relative to the naive policy—and it still wins (cut from this subsection 08-12)]

[1]Calibration quality matters and is itself a finding. The conditioned policies in this paper are separately trained instances of the same design: the robot picks (Sec. V-D) and the replication pair quoted in Sec. V-B use an earlier calibration; the recipe ablation and the live counterfactuals (Sec. V-E) use offsets re-estimated from the measured contact distribution plus the unsaturated $\hat{c}_{\mathrm{fmag}}$ channel.

[2]Injecting at the action-expert input instead acquired consistently less authority in preliminary experiments.

C. Counterfactual probe battery

All probes intervene on a trained, frozen policy; none involves retraining. The offline probes (P1–P4) run on held-out validation episodes. We define **authority** as the intervention-induced change in the commanded vertical action on committed-descent frames, expressed as a percentage of the mean commanded descent: 0% means the intervention is ignored, 100% means it fully cancels descent, >100% means reversal (retreat).

(P1) Condition forcing (open loop). With images and state frozen, $\hat{c}$ is forced from its observed value to a target pattern—either all channels saturated (*full forcing*) or a calibrated contact-onset pattern matching measured first-contact statistics (*contact-moment forcing*). Applicable to conditioned models only.

(P2) State transplant (open loop). The force-bearing dimensions of a *measured* first-contact state are transplanted into pre-contact descent frames. This probe applies to any policy, including the naive policy—it is what puts both policies on the same scale. Routing the transplant selectively, to the $\hat{c}$ computation alone (d_{FiLM}) or the raw state alone (d_{raw}), attributes the response to each route. [TODO: verify exact transplanted dims (wrench+seal vs full state) in probe_state_authority.py before submission]

(P3) Dose-response sweep. The transplanted wrench is rescaled to target magnitudes (8, 10, 12 N), tracing the policy's response as a function of force *dose*. The demonstrations peak near the low end of this range, so the sweep measures extrapolation: a policy that has bound braking to force should brake *more* at higher dose; one that has memorized a contact template should respond most near the template and fade beyond it. The sweep's *slope sign* is the paper's central discriminator.

(P4) Closed-loop press simulation. Open-loop probes cannot show consequences. We place the frozen policy in a closed loop at each episode's nominal first-contact timing: commanded descent advances a virtual penetration depth, a linear-stiffness force model raises the simulated wrench accordingly, and the seal event is withheld—simulating the operationally critical misalignment scenario in which no press depth ever produces a seal. Outcomes: whether the policy stops before the force limit, and maximum penetration. A control variant grants the seal at a shallow depth, separating force-driven stopping from seal-event-driven stopping.

(P5) Live counterfactual probe. The full loop contains physics no offline probe reproduces. On the robot, we freeze observations at staged poses along the approach and apply (P1)–(P3) interventions to *two* policies—the conditioned policy and the naive policy—on identical frozen observations, so both receive the same physical counterfactual.

Scope. P1–P3 are single-frame and open-loop; P4 closes the loop under a force model; P5 closes the gap to the real system. No single probe is decisive—the findings below rest on their agreement.

[TODO: dose-response figure: cancellation vs. forced dose (8/10/12 N) for the four ablation policies — assets: vla/probes/0729_state_*_ramp*.txt]

Fig. 2. Dose-response under forced wrench magnitude (P3). The naive policy's response fades past the demonstrated range; the unmasked variant lies on top of it (all response through the raw route); the ungrounded pathway is flat at zero; only the grounded pathway grows with dose, crossing full cancellation into commanded retreat at 12 N.

V. Experiments

The offline experiments compare four policies that start from the *same* trained naive checkpoint and continue training on the *same* demonstrations: the naive policy itself, the full recipe (the conditioned policy), and two controls that each remove a single ingredient—the mask or the grounding (Sec. V-B). Imitation error on held-out episodes is indistinguishable across all four (0.84–0.96 mm per step): on the data manifold, these are the same policy. Every reported model is its validation-loss-selected (*best*) checkpoint—one uniform selection rule for all policies—and the robot deploys the same checkpoints.

A. The two policies under intervention

We walk the probe battery in order: P1–P4 against the naive and conditioned policies here (Table I, Fig. 2), the ablation controls in Sec. V-B, and the closed-loop picks and the live probe P5 on the robot (Sec. V-D–V-E).

(P1) Condition forcing. Forcing every channel of $\hat{c}$ to saturation moves the conditioned policy's committed descent by $+0.9$ mm per frame. Restricting the forcing to a calibrated contact-onset pattern shows the limit of what any policy adopts: even the best-conditioned policy moves by only 7%—the 1–2-frame contact *transient* is weakly adopted across the board, a point Sec. VI returns to.

(P2) State transplant: both policies on one scale. Transplanting a measured first-contact state into pre-contact descent frames moves the naive policy by $+1.63$ mm—105% of its descent command. The naive policy is not force-blind: its raw route responds strongly to force-bearing state. In the conditioned policy the same transplant lands entirely on d_{FiLM} ($+1.57$ mm, 94%; the masked raw route contributes nothing)—the mask working as designed. *What* is transplanted matters as much as whether: a *sealed* state moves both models far more than the contact state does (on-robot: 147% vs. 67% for the naive policy; Sec. V-E). Together with the weak contact-onset forcing above, the pattern is that stable, persistent conditions (seal, sustained press) are adopted essentially for free, while a transient that is redundant with depth-like cues in ordinary demonstrations is not. The operational corollary is developed in Sec. VI: on stiff contact, even substantial condition authority cannot succeed by *reactive* gating at the contact moment—the viable mechanism is an anticipatory behavior prior plus post-hoc state gating, which is exactly what the probes measure as adopted.

(P3) Dose-response: raw access binds a template, the grounded pathway binds a brake. The naive policy responds to transplanted force—at the demonstrated magnitude. As the dose rises past the demonstrations, its response *fades* (36% → 28%): a template, matched ever more poorly as reality departs from it. The conditioned policy's response *grows* with dose, crossing full cancellation into commanded retreat at 12 N (31% → 115%)—where the unsaturated channel reads five times its training range. The monotone extrapolation is not learned from data (no demonstration presses at these forces); it is inherited from the channel's design as a calibrated monotone scalar. The forms are not artifacts of checkpoint selection: no naive checkpoint we examined exceeds 65% cancellation at any dose, while the grounded brake remains monotone to the end of training.

(P4) Press simulation: consequences. Closing the loop under the seal-never press turns the response forms into outcomes. Every conditioned episode settles by 9.8 mm of penetration, with contact force at or below 7.1 N when the simulation ends. The naive policy settles in four of six episodes but runs away in two—still descending when the step budget ends, at up to 14.9 mm with force climbing through 13.1 N; its later checkpoints respond *more* in open loop yet fare *worse* here (3/6, max 16.0 mm), because a response that saturates below full cancellation cannot stop a stiff press regardless of its size. The seal-granted control variant localizes the failure: given a seal event at shallow depth the naive policy stops normally (0.5 mm over-travel)—withhold the seal and it presses through. Its stopping is real but bound to the *seal event*; the conditioned policy's is bound to *force*.

B. Ablating the recipe: the mask and the grounding

The two controls run the same battery (Table II); each isolates one ingredient.

Remove the mask, and force reroutes: bypass. The unmasked control keeps the raw wrench in the state alongside $\hat{c}$, and behaves as the naive policy everywhere: its dose-response curve lies on top of the naive policy's ($+1.57 \rightarrow +1.12$ vs. $+1.41 \rightarrow +1.12$; identical at 12 N), forcing its $\hat{c}$ moves nothing (0.0 mm), and the pathway decomposition places its transplant response entirely on the raw route (d_{FiLM}=0.04). With a redundant raw route available, gradient descent satisfies the imitation objective through the already-functional path and starves the dedicated pathway. This is the *bypass* of Sec. I—and nothing in training registers it: the unmasked variant's validation loss matches the conditioned policy's to the fourth decimal (0.14750 vs. 0.14762). An independently trained pair under the earlier calibration reproduces the result: 7% condition-forcing authority with the wrench kept versus 76% with it masked, again at near-identical loss. [MEMO: PARKED (08-12): de-emphasize the loss axis. Remaining loss artifacts: the fourth-decimal phrasing here and in the abstract, "unchanged validation loss" in Sec. I para. 4, and "indistinguishable in loss" in the closing one-sentence summary. Candidate replacement: held-out imitation error (0.85 vs. 0.87 mm/step for this pair).]

Remove the grounding, and the pathway is worse than raw access. The shuffled control has the same added param-

TABLE I

THE NAIVE AND CONDITIONED POLICIES UNDER THE OFFLINE BATTERY. ALL RESPONSES IN MM/FRAME OF COMMANDED DESCENT; *own* = THE POLICY'S MEAN COMMANDED DESCENT ON THE EVALUATED FRAMES, THE DENOMINATOR OF THE PERCENTAGE. FORCING (P1) = EVERY $\hat{c}$ CHANNEL FORCED TO SATURATION, ON COMMITTED-DESCENT FRAMES; THE NAIVE POLICY HAS NO $\hat{c}$ TO FORCE (—). TRANSPLANT (P2) = RESPONSE TO A TRANSPLANTED FIRST-CONTACT STATE. DOSE-RESPONSE (P3) = RESPONSE AT FORCED 8 N→12 N. PRESS SIM (P4) = MAXIMUM PENETRATION IN THE SEAL-NEVER PRESS SIMULATION.

Policy	Force access	Forcing (P1)	Transplant (P2)	Dose-response (P3)	Press sim (P4)
naive VLA	raw wrench	—	$+1.63$ (105%) own -1.55	$+1.41 \rightarrow +1.12$ (36→28%) ↓ own -3.94	max 14.9 mm
conditioned policy	grounded pathway	$+$**0.9** own -7.8	$+1.57$ (94%) own -1.67	$+1.20 \rightarrow +$**4.40** (31→**115%**) ↑ own -3.84	**max 9.8 mm**

TABLE II

ABLATION CONTROLS: SAME INITIALIZATION, SAME DATA, ONE INGREDIENT REMOVED. COLUMNS AS IN TABLE I.

Policy	Force access	Forcing (P1)	Transplant (P2)	Dose-response (P3)	Press sim (P4)
FiLM, wrench kept	raw + dead $\hat{c}$	0.0 own -7.7	$+1.68$ own -1.52	$+1.57 \rightarrow +1.12$ own -3.77	max 11.9 mm
FiLM, mask, shuffled $\hat{c}$	ungrounded pathway	-0.2 own -8.2	-0.06 (0%) own -1.35	$-0.06 \rightarrow -0.11$ ($\approx$0) own -1.35	**max 57.2 mm**

eters, the same continued training, the same pathway—and zero response at every dose (-0.2 mm under forcing; flat dose-response), with the worst closed-loop outcome in the study: no episode settles, all six still descending at the step budget, 19–23 N past the force limit, up to 57 mm deep. Grounding, not capacity, is the active ingredient: the mask removed the raw route, and the noise-trained $\hat{c}$ replaced it with nothing.

C. Scale does not change the class

Fine-tuning π_0 [24] and $\pi_{0.5}$ [25] (3B-class, $\sim 8\times$ SmolVLA) on the same demonstrations reproduces the failure at scale, in two forms. $\pi_{0.5}$ shows the template signature outright: a strong on-manifold transplant response (65%) whose dose-response saturates shallowly (17% → 25%). π_0's response rises with dose (20% → 80%) yet stays below full cancellation—and below what stopping a stiff press requires. Both collapse in the seal-never press: no episode settles, all still descending when the step budget ends, at max penetrations of 48 and 19.6 mm (versus 14.9 for the naive SmolVLA, and 9.8, settled, for the conditioned policy), at comparable imitation error (0.89–1.03 mm per step). Neither parameter count nor architecture converts raw force access into force-bound braking; what changes the class is the conditioning pathway, not scale.

D. Robot validation: closed-loop picks

The offline battery makes three predictions about the real closed loop. A template response that fades beyond the demonstrations should let contact force run away until something outside the policy intervenes; a monotone brake should stop on its own, at forces near the demonstrated range; and the two responses should cross between the demonstrated magnitudes and the safety limit. The robot experiments test these predictions, and the outcome variable is the *force behavior* of each attempt—task success follows from it by definition.

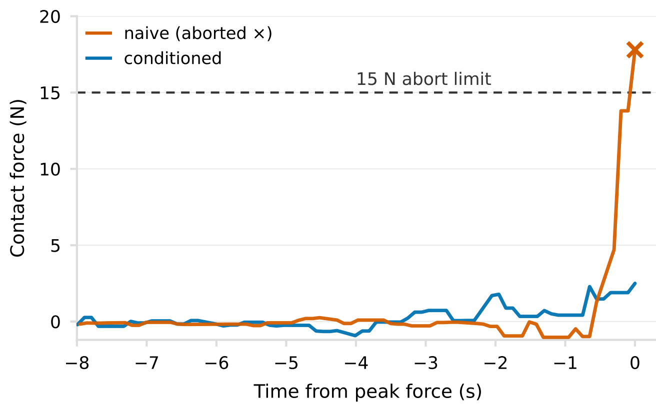

Fig. 3. Contact force on the robot: one representative trial per policy, aligned at peak force (t=0), over the interaction window (before lift, where carrying the case would re-enter payload weight into the measured force). The naive attempt rises through the demonstrated range and is terminated by the external 15 N abort (×; all three naive attempts end this way, at 15.4–18.7 N), while the conditioned pick presses to 2.5 N—the median of its nine picks across three stack heights, one of them absent from the demonstrations—and stops on its own.

The predictions hold (Fig. 3). Under the naive policy every attempt ends the same way: contact, continued descent, and an external force-limit abort at 15.4–18.7 N—the closed-loop realization of a response that saturates below full cancellation. The conditioned policy stops itself in *every* pick, across three stack heights: 3/3 at the naive policy's own height, and 3/3 at an intermediate height absent from the demonstrations, which contain only the other two. Peak force, measured over the *interaction window* (descent through retreat, before lifting re-enters payload weight into the wrench; the abort monitors the same quantity), stays at or below 3.1 N in seven of the nine picks (median 2.5 N; two picks sealed with no measurable compression at all). The two force distributions do not overlap, [TODO: live counterfactual figure: fair-dose bars + high-dose crossover curves (assets: vla/live_film_probes 08-06 runs, _vs_naive.png)]

Fig. 4. On-robot counterfactuals (P5), naive vs. recalibrated conditioned policy on identical frozen observations. Left: at demonstration-level doses the naive response is equal or larger. Right: raising the dose to $+6/+9/+12$ N reverses the ordering between 6 and 9 N—the offline crossover, reproduced on the physical system.

and the task outcomes are their threshold readout: $0/3$ versus $9/9$. The two harder presses (10.2 and 14.5 N) are the brake visible near its limit—both stopped short of the abort threshold and completed the pick where the naive policy, at the same forces, was still commanding descent.

The stopping runs through the conditioning pathway. The deployed policy's own dose-response is monotone: transplanted wrenches at $8/10/12$ N cancel $3\% \rightarrow 7\% \rightarrow 13\%$ of committed descent—rising with dose, and carried entirely by the FiLM path (the masked raw route contributes $+0.00$ mm, the mask working as designed). During the picks themselves the modulation is measurably active at the transitions ($|\gamma|$ $+48\%$, $|\beta|$ $+79\%$ at contact/seal relative to descent), and forcing $\hat{c}$ at ten frozen poses on the robot moves the commanded action in the correct direction at every staged condition (contact $+0.73$, sealed $+2.09$, $+6$ N $+2.18$ mm/frame).

E. Robot validation: on-robot counterfactuals

The live probe (P5) freezes observations at staged poses and doses the naive policy and the conditioned policy with identical counterfactual states. Two sessions matter (Fig. 4).

Fair dose. At transplant doses matching the demonstrations ($\leq$ contact magnitude), the *naive* policy responds as strongly or stronger than the conditioned policy (first-contact: 67% vs. 32% of own descent; sealed state: 147% vs. 58%)—in quantitative agreement with the offline dose-response curves at the same doses, where the template peaks. The conditioned policy's response is dose-monotone across the staged conditions (hover $+0.29$ $\rightarrow$ pre-seal $+0.87$ $\rightarrow$ sealed $+1.41$ $\rightarrow$ $+6$ N $+1.70$ mm/frame).

High dose. Raising the forced normal force to $+6/+9/+12$ N reproduces the offline crossover on the robot: the conditioned policy accelerates its braking, $+1.19 \rightarrow +4.59 \rightarrow +8.56$ mm/frame—at 12 N, twice its own descent command, i.e., commanded retreat—while the naive policy's response collapses at the margin ($+1.34$ $\rightarrow$ $+3.18$ $\rightarrow$ $+4.18$; increments $+1.84 \rightarrow +1.00$). The reversal sits between 6 and 9 N. At near-surface poses the separation is starkest: naive $+1.0$–1.8 vs. conditioned $+7.5$ mm/frame at 12 N.

Isn't the naive policy better, then? At mid dose, yes—and it does not matter. Over-press is a dynamical process in which force rises *past* the demonstrated magnitude; what determines the outcome is the response exactly where the template fades. The naive policy's brake weakens as it becomes more necessary (closed loop: force runaway in every robot attempt; press-simulation episodes that never settle), while the conditioned brake strengthens (self-stop in every pick; every simulated press settling). The naive policy's strongest counterfactual response is to the *sealed* state—but over-press is precisely the scenario in which the seal never arrives: its most powerful brake is bound to an absent signal. Similar in loss, similar at the demonstrated doses, categorically different where the task is decided.

Three validity notes. Force-torque baselines were re-anchored at a pre-descent hover before each probe session, and absolute transplant targets were drift-shifted so both policies received the same physical dose; live conditioned-policy magnitudes are conservative lower bounds, as some frozen states are slightly off-manifold for its input distribution. The naive policy's closed-loop attempts were run at a single stack height—its failure mode is uniform and force-driven, but the height-generalization evidence applies to the conditioned policy only. High-dose probes cover three poses; we report them as targeted supplementary measurements, not a survey.

VI. Discussion and Limitations

Why the prescription must be anticipatory. A first-principles calculation explains both the failure and the fix. The case stack is stiff: $\sim$5 N/mm of penetration. Stopping below the 15 N limit therefore requires a full stop within $\sim$2 mm of contact—under 330 ms action chunks and the demonstrated descent rates, this is arithmetically impossible for *reactive* gating that begins braking at the contact transient, no matter how much authority the transient carries. The viable mechanisms are anticipatory: approach slowly, press deliberately, retreat. The press-retreat demonstrations teach exactly this behavior prior, and the probes confirm the resulting division of labor: the successful policy's contact-moment authority is small (7%) while its full-condition authority is large (76%)—success comes from an anticipatory behavior prior plus post-hoc state gating, not from a reflex. The instrument's value here is not that it found a large number, but that it correctly *predicted which mechanism could not work*.

Operational asymmetry. The computed condition creates knobs the baseline lacks. Force-torque baselines drift ($\sim$1 N within a day, pose-sensitive); the conditioned policy re-anchors its offsets at a hover pose each run—a deployment-time correction with no retraining—while the baseline's force statistics are frozen in its normalization. The same interface is what makes the policy auditable (P1/P5 probes) and intervenable at deployment.

Limitations. Single task, single platform, small robot-n: the robot results are a case study, and the statistical claims live in the offline battery (n=6 held-out episodes $\times$ dense frames, six policies across three architectures). The press simulation depends on a linear force model; we mitigate with the seal-granted control and the live probes, which bracket the simulated mechanism from both sides. Contact-moment (reactive) authority plateaus low in every configuration we trained—in development phases, oversampling transition frames also failed to move it; distilling an explicit descend-until-contact oracle (e.g., via DAgger [20], [26]) is the natural next step, as is a negative-control task whose critical condition is *not* observable in the state, and replication of the bypass audit on force-fusion architectures whose designs assume use [1], [3].

VII. Conclusion

On a task whose critical transition is observable only through force, we showed that a behavior-cloned VLA policy given the full signal does not bind its behavior to it—and that no number in the standard validation stack could have revealed this. A computed condition vector behind a zero-new-information FiLM pathway turns the question into a measurement: counterfactual authority probes expose bypass at identical loss, localize authority to stable post-hoc states rather than the contact transient, and—in a matched quintuple with indistinguishable imitation error—show that the data's correlation structure decides *whether* force is used while the conditioning design decides the *form* its use takes: a template that fades beyond the demonstrations, or a calibrated brake that strengthens monotonically, stops, and retreats. The distinction is invisible on the manifold and decisive off it, on the robot as offline. Imitation pins down the trajectory. If we want the mechanism pinned too, we must give the policy demonstrations that decorrelate the condition from its correlates—and an interface through which the condition's use can be shaped, measured, and corrected.

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